

A.1.4. Direct Preference Optimization (DPO)

The objective of DPO is:

$$\mathcal{J}_{DPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o^+, o^- \sim \pi_{sft}(O|q)] \log \sigma \left(\beta \frac{1}{|o^+|} \sum_{t=1}^{|o^+|} \log \frac{\pi_{\theta}(o_t^+|q, o_{<t}^+)}{\pi_{ref}(o_t^+|q, o_{<t}^+)} - \beta \frac{1}{|o^-|} \sum_{t=1}^{|o^-|} \log \frac{\pi_{\theta}(o_t^-|q, o_{<t}^-)}{\pi_{ref}(o_t^-|q, o_{<t}^-)} \right) \quad (12)$$

The gradient of $\mathcal{J}_{DPO}(\theta)$ is:

$$\begin{aligned} \nabla_{\theta} \mathcal{J}_{DPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o^+, o^- \sim \pi_{sft}(O|q)] & \left(\frac{1}{|o^+|} \sum_{t=1}^{|o^+|} GC_{DPO}(q, o, t) \nabla_{\theta} \log \pi_{\theta}(o_t^+|q, o_{<t}^+) \right. \\ & \left. - \frac{1}{|o^-|} \sum_{t=1}^{|o^-|} GC_{DPO}(q, o, t) \nabla_{\theta} \log \pi_{\theta}(o_t^-|q, o_{<t}^-) \right) \end{aligned} \quad (13)$$

Data Source: question in SFT dataset with outputs sampled from SFT model. Reward Function: human preference in the general domain (can be 'Rule' in mathematical tasks). Gradient Coefficient:

$$GC_{DPO}(q, o, t) = \sigma \left(\beta \log \frac{\pi_{\theta}(o_t^-|q, o_{<t}^-)}{\pi_{ref}(o_t^-|q, o_{<t}^-)} - \beta \log \frac{\pi_{\theta}(o_t^+|q, o_{<t}^+)}{\pi_{ref}(o_t^+|q, o_{<t}^+)} \right) \quad (14)$$

A.1.5. Proximal Policy Optimization (PPO)

The objective of PPO is:

$$\mathcal{J}_{PPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o \sim \pi_{\theta_{old}}(O|q)] \frac{1}{|o|} \sum_{t=1}^{|o|} \min \left[\frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})} A_t, \text{clip} \left(\frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})}, 1 - \epsilon, 1 + \epsilon \right) A_t \right]. \quad (15)$$

To simplify the analysis, it is assumed that the model only has a single update following each exploration stage, thereby ensuring that $\pi_{\theta_{old}} = \pi_{\theta}$. In this case, we can remove the min and clip operation:

$$\mathcal{J}_{PPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o \sim \pi_{\theta_{old}}(O|q)] \frac{1}{|o|} \sum_{t=1}^{|o|} \frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{old}}(o_t|q, o_{<t})} A_t. \quad (16)$$

The gradient of $\mathcal{J}_{PPO}(\theta)$ is:

$$\nabla_{\theta} \mathcal{J}_{PPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), o \sim \pi_{\theta_{old}}(O|q)] \frac{1}{|o|} \sum_{t=1}^{|o|} A_t \nabla_{\theta} \log \pi_{\theta}(o_t|q, o_{<t}) \quad (17)$$

Data Source: question in SFT dataset with outputs sampled from policy model. Reward Function: reward model. Gradient Coefficient:

$$GC_{PPO}(q, o, t, \pi_{\theta_{rm}}) = A_t, \quad (18)$$

where A_t is the advantage, which is computed by applying Generalized Advantage Estimation (GAE) (Schulman et al., 2015), based on the rewards $\{r_{\geq t}\}$ and a learned value function V_{ψ} .

A.1.6. Group Relative Policy Optimization (GRPO)

The objective of GRPO is (assume $\pi_{\theta_{old}} = \pi_{\theta}$ for simplified analysis):

$$\begin{aligned} \mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P_{sft}(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)] \\ \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left[\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{old}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t} - \beta \left(\frac{\pi_{ref}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} - \log \frac{\pi_{ref}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} - 1 \right) \right]. \end{aligned} \quad (19)$$

The gradient of $\mathcal{J}_{GRPO}(\theta)$ is:

$$\begin{aligned} \nabla_{\theta} \mathcal{J}_{GRPO}(\theta) &= \mathbb{E}[q \sim P_{sft}(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)] \\ &\quad \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left[\hat{A}_{i,t} + \beta \left(\frac{\pi_{ref}(o_{i,t}|o_{i,<t})}{\pi_{\theta}(o_{i,t}|o_{i,<t})} - 1 \right) \right] \nabla_{\theta} \log \pi_{\theta}(o_{i,t}|q, o_{i,<t}). \end{aligned} \quad (20)$$

Data Source: question in SFT dataset with outputs sampled from policy model. Reward Function: reward model. Gradient Coefficient:

$$GC_{GRPO}(q, o, t, \pi_{\theta_{rm}}) = \hat{A}_{i,t} + \beta \left(\frac{\pi_{ref}(o_{i,t}|o_{i,<t})}{\pi_{\theta}(o_{i,t}|o_{i,<t})} - 1 \right), \quad (21)$$

where $\hat{A}_{i,t}$ is computed based on the group reward scores.